

JAVA PROGRAMMING

Find the largest element and smallest element in the Array

```
import java.util.Scanner;

public class Main{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements in the array: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.print("Enter the elements of the array: ");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        int max = arr[0];

        int min = arr[0];

        for (int i = 1; i < n; i++) {

            if (arr[i] > max) {

                max = arr[i];

            }

            if (arr[i] < min) {

                min = arr[i];

            }

        }

        System.out.println("The largest element in the array is: " + max);
```

```
        System.out.println("The smallest element in the array is: " + min);
    }
}
```

Find the sum of the positive elements in the Array

```
import java.util.Scanner;

public class Main{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int sum = 0;

        System.out.print("Enter the number of elements in the array: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.print("Enter the elements of the array: ");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

            if (arr[i] > 0) {

                sum += arr[i];

            }

        }

        System.out.print("The sum of the positive elements in the array
is: " + sum);
    }
}
```

```
}  
}
```

Sum of two Array problem

```
import java.util.Scanner;  
  
public class Main{  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter the number of elements in the array: ");  
        int n = sc.nextInt();  
        int[] arr = new int[n];  
        System.out.print("Enter the elements of the array: ");  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
        System.out.print("Enter the target sum: ");  
        int target = sc.nextInt();  
        for (int i = 0; i < n; i++) {  
            for (int j = i + 1; j < n; j++) {  
                if (arr[i] + arr[j] == target) {  
                    System.out.println("The two numbers that add up to " + target + "  
are: " + arr[i] + " and " + arr[j]);  
                }  
            }  
        }  
    }  
}
```

Odd to Even and Even to Odd count

```
import java.util.Scanner;

public class Main{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int oddToEvenCount = 0;

        int evenToOddCount = 0;

        System.out.print("Enter the number of elements in the array: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.print("Enter the elements of the array: ");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        for (int i = 0; i < n - 1; i++) {

            if (arr[i] % 2 != 0 && arr[i + 1] % 2 == 0) {

                oddToEvenCount++;

            }

            if (arr[i] % 2 == 0 && arr[i + 1] % 2 != 0) {

                evenToOddCount++;

            }

        }

        System.out.println("The count of odd to even transitions is: " +
oddToEvenCount);
```

```
        System.out.println("The count of even to odd transitions is: " +  
evenToOddCount);  
    }  
}
```

Search an element in the Array

```
import java.util.Scanner;  
  
public class Main{  
    public static void main(String []args){  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter the number of elements in the array: ");  
  
        int n = sc.nextInt();  
  
        int[] arr = new int[n];  
  
        System.out.print("Enter the elements of the array: ");  
  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        System.out.print("Enter the element to search: ");  
  
        int target = sc.nextInt();  
  
        int count = 0;  
  
        for (int i = 0; i < n; i++) {  
            if (arr[i] == target) {  
                count++;  
            }  
        }  
  
        if (count > 0) {  
            System.out.println("The element " + target + " is present in the array and it  
occurs " + count + " times.");  
        }  
    }  
}
```

```
    } else {  
        System.out.println("The element " + target + " is not present in the array.");  
    }  
}
```

Reverse an Array

```
import java.util.Scanner;  
  
public class Main{  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter the number of elements in the  
array: ");  
  
        int n = sc.nextInt();  
  
        int[] arr = new int[n];  
  
        System.out.print("Enter the elements of the array: ");  
        for (int i = 0; i < n; i++)  
        {  
            arr[i] = sc.nextInt();  
        }  
  
        System.out.println("\nThe reversed array is: ");  
        for (int i = n - 1; i >= 0; i--)  
        {  
            System.out.print(arr[i] + " ");  
        }  
    }  
}
```

```
}  
  
}
```

Frequency of elements in the Array

```
import java.util.Scanner;  
  
public class Main{  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter the number of elements in the array: ");  
  
        int n = sc.nextInt();  
  
        int[] arr = new int[n];  
  
        System.out.print("Enter the elements of the array: ");  
  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        System.out.println("\nThe frequency of elements in the array is: ");  
  
        for (int i = 0; i < n; i++) {  
            int count = 1;  
  
            for (int j = i + 1; j < n; j++) {  
                if (arr[i] == arr[j]) {  
                    count++;  
  
                    arr[j] = Integer.MIN_VALUE; // Mark as counted  
                }  
            }  
  
            if (arr[i] != Integer.MIN_VALUE) {
```

```

        System.out.println(arr[i] + " occurs " + count + " times.");
    }
}
}}
```

Remove duplicates from the Array

```

import java.util.Scanner;

public class Main{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements in the array: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.print("Enter the elements of the array: ");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        System.out.println("\nThe array after removing duplicates is: ");

        for (int i = 0; i < n; i++) {

            int count = 0;

            for (int j = 0; j < i; j++) {

                if (arr[i] == arr[j]) {

                    count++;

                }

            }

            if (count == 0) {

                System.out.print(arr[i] + " ");

            }

        }

    }

}
```



```
    }  
  }  
}
```

Adjacent stick Game

```
import java.util.*;  
  
public class Main{  
    public static void main(String [] args){  
        Scanner sc=new Scanner(System.in);  
        System.out.print("Enter Size of the array1: ");  
        int n1=sc.nextInt();  
        System.out.print("Enter array1 values: ");  
        int arr1[]=new int[n1];  
        for(int i=0;i<n1;i++)  
        {  
            arr1[i]=sc.nextInt();  
        }  
  
        System.out.print("Enter Size of the array2: ");  
        int n2=sc.nextInt();  
        System.out.print("Enter array2 values: ");  
        int arr2[]=new int[n1];
```

```
        for(int i=0;i<n2;i++)
        {
            arr2[i]=sc.nextInt();
        }

        int mergedArr[]=new int[n1+n2];
        for(int i=0;i<n1;i++)
        {
            mergedArr[i]=arr1[i];
        }

        for(int i=0;i<n2;i++)
        {
            mergedArr[n1+i]=arr2[i];
        }

        System.out.println("Merged array: ");
        for(int x:mergedArr)
            System.out.print(x+" ");
    }
}
```

Find the array elements which sum is the Target

```
import java.util.*;

public class Main{

    public static void main(String [] args){

        Scanner sc=new Scanner(System.in);

        System.out.print("Enter Size of the array: ");

        int n=sc.nextInt();

        System.out.print("Enter array values: ");

        int arr[]=new int[n];

        for(int i=0;i<n;i++)

            arr[i]=sc.nextInt();

        System.out.print("Enter the target sum: ");

        int target=sc.nextInt();

        for(int i=0;i<n;i++){

            for(int j=i+1;j<n;j++){

                if(arr[i]+arr[j]==target){

                    System.out.println("The two numbers that add up to " + target

+ " are: " + arr[i] + " and " + arr[j]);

                }

            }

        }

    }

}
```

```
    }  
  }  
}
```

You are given an integer array a of size n which represents houses from i th house you can go directly to $(i+a[i])$ th house. if it is present. your task is to find the minimum number of jumps required to reach the last house from the first house. if it is not possible to reach the last house return -1

```
import java.util.*;  
  
public class Main{  
    public static void main(String [] args){  
        Scanner sc=new Scanner(System.in);  
        System.out.print("Enter Size of the array: ");  
        int n=sc.nextInt();  
        System.out.print("Enter array values: ");  
        int arr[]=new int[n];  
        for(int i=0;i<n;i++){  
            arr[i]=sc.nextInt();  
        }  
        int visited=0;  
        int currentEnd=0;  
        int farthest=0;  
        for(int i=0;i<n-1;i++){  
            farthest=Math.max(farthest,i+arr[i]);  
            if(i==currentEnd){
```

```

        visited++;
        currentEnd=farthest;
        if(currentEnd>=n-1)
            break;
    }
}
if(currentEnd<n-1)
    System.out.println(-1);
else
    System.out.println(visited);
}
}

```

By this code Print A and B in a single Line :

```

import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter two long integers: ");

        long a = sc.nextLong();

        long b = sc.nextLong();

        System.out.println(a + " " + b);
    }
}

```

```
}  
}
```

Round of two decimal places using Float

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a decimal number: ");  
        double num = sc.nextDouble();  
        System.out.printf("Rounded number: %.2f", num);  
    }  
}
```

Round-off a number to the nearest integer using Math ceil and floor

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a decimal number: ");  
        double num = sc.nextDouble();  
        System.out.println("Rounded up (ceiling): " + Math.ceil(num));  
    }  
}
```

```
        System.out.println("Rounded down (floor): " +  
Math.floor(num));  
        System.out.println("Rounded to nearest integer: " + (int)(num));  
    }  
}
```

BY THE GIVEN ARRAY TAKE 2 NO TO SUM AND GET SAME OUTPUT PAIR

```
import java.util.*;  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int[] a = new int[n];  
  
        for (int i = 0; i < n; i++) {  
            a[i] = sc.nextInt();  
        }  
        int t = sc.nextInt();  
        Arrays.sort(a);  
        int l = 0;  
        int r = n-1;  
        int s = a[l]+a[r];
```

```

while(l < r){
    if(s==t){
        System.out.print("(" + a[l] + ", " + a[r] + ")");
        l++;
        r--;
    }
    else if(s<t){
        l++;
    }
    else{
        r--;
    }
}
}
}
}

```

**BY THE GIVEN ARRAY TAKE 3 NO TO SUM
AND GET SAME OUTPUT PAIR**

```

import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

```



```

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int[] a1 = new int[n];


for (int i = 0; i < n; i++) {
    a1[i] = sc.nextInt();
}

int t = sc.nextInt();

for(int i=0;i<n;i++){
    for(int j=i+1;j<n;j++){
        for(int k=j+1;k<n;k++){
            if(a1[i]+a1[j]+a1[k]==t){
                System.out.print("(" + a1[i] + "," + a1[j] + "," + a1[k] + ")");
            }
        }
    }
}

}

}

}

```

MATH FUNCTIONS

MATH.ABS - NEGATIVE TO POSITIVE

MATH.SQRT - GIVE THE SQUARE ROOT VALUE

MATH.POW(2,3)-IT TAKE THE VALUE OF 2^3

MATH.MAX-FIND THE MAXIMUM NUMBER

MATH.MIN - FIND THE MINIMUM NUMBER

MATH.CEIL – ROUND THE DECIMAL TO WHOLE INTEGER

MATH.FLOOR - ROUND OFF THE LOWEST VALUE

MATH.ROUND - ROUND THE VALUES

ADD TWO ARRAY WITH THE EACH INDEX AND GIVE THE OUTPUT

```
import java.util.*;  
  
public class Main {  
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int a = sc.nextInt();
```

```
int[] a1 = new int[a];
```

```
for (int i = 0; i < a; i++) {
```

```
    a1[i] = sc.nextInt();
```

```
}
```

```
int b = sc.nextInt();
```

```
int[] a2 = new int[b];
```

```
for (int i = 0; i < b; i++) {
```

```
    a2[i] = sc.nextInt();
```

```
}
```

```
int[] a3=new int[a];
```

```
for(int i=0;i<a;i++){
```

```
    a3[i]=a1[i]+a2[i];
```

```
}
```

```
Arrays.sort(a3);
```

```
for(int i=0;i<a;i++){
```

```
    System.out.print(a3[i]+" ");
```

```
}
```

```
}  
}
```

Addition of two different size arrays element wise

```
import java.util.*;  
  
public class Main{  
    public static void main(String[] args) {  
        Scanner sc=new Scanner(System.in);  
        System.out.print("Enter size of array1: ");  
        int n1=sc.nextInt();  
        int a[]=new int[n1];  
        System.out.print("Enter array1 values: ");  
        for(int i=0;i<n1;i++)  
            a[i]=sc.nextInt();  
        System.out.print("Enter size of array2: ");  
        int n2=sc.nextInt();  
        int b[]=new int[n2];  
        System.out.print("Enter array2 values: ");  
        for(int i=0;i<n2;i++)  
            b[i]=sc.nextInt();  
        int maxSize=Math.max(n1, n2);  
        int c[]=new int[maxSize];
```

```

System.out.print("The result array: ");
for(int i=0;i<maxSize;i++){
    int valA=(i<n1)?a[i]:0;
    int valB=(i<n2)?b[i]:0;
    c[i]=valA+valB;
    System.out.print(c[i]+" ");
}
}
}

```

Shift zeros to the end of the Array

```

import java.util.*;

public class Main{

    public static void main(String [] args){

        Scanner sc=new Scanner(System.in);

        System.out.print("Enter array size: ");

        int n=sc.nextInt();

        int a[]=new int[n];

        System.out.print("Enter array values: ");

        for(int i=0;i<n;a[i]=sc.nextInt(),i++);

        for(int i=0;i<n;i++){

            if(a[i]!=0){

                System.out.print(a[i]+" ");

            }

        }

    }

}

```

```

    }
    for(int i=0;i<n;i++){
        if(a[i]==0){
            System.out.print(a[i]+" ");
        }
    }
}
}
}

```

Convert upper to lower using ternary operator

```

import java.util.*;

public class Main{
    public static void main(String []args){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a character: ");
        char a=sc.next().charAt(0);
        int no=(int)a;
        System.out.println(no>96?(char)(no-32):(char)(no+32));
    }
}

```

Palindrome string

```

import java.util.*;

public class Main{

```

```

public static void main(String[] args) {
    Scanner sc=new Scanner(System.in);
    System.out.println("Enter a string: ");
    String org=sc.next();
    String rev="";
    for(int i=org.length()-1;i>=0;i--)
        rev+=org.charAt(i);

    System.out.println(org.equalsIgnoreCase(rev)?org+ " is a
palindrome":org+" is not a palindrome");
}
}

```

TO GET THE ASCII VALUE NUMBER

```

import java.util.*;

public interface Main {

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter the character: ");
        char a=sc.next().charAt(0);
        System.out.println("ASCII value of "+a+" is "+(int)a);
    }
}

```

Rating app Problem

```

import java.util.*;

public class Main{

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        int rating[]=new int[5];

        int sum=0;

        for(int i=0;i<5;i++){

            System.out.print("Enter rating (1-5) of the product"+(i+1)+":
");

            rating[i]=sc.nextInt();

            while(rating[i]<=0||rating[i]>5){

                System.out.println("Invalid rating! Rate between(0-5)");

                System.out.print("Enter rating (1-5) of the
product"+(i+1)+": ");

                rating[i]=sc.nextInt();

            }

        }

        for(int x:rating){

            sum+=x;

        }

        System.out.println("Average rating: "+(float)sum/5);

        System.out.println((float)sum/5>3.8?"Discount available!!":"No
discount! Try next time.");

    }

}

```


Number of vowels

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a word: ");

        String word = sc.nextLine().toLowerCase();

        int n = word.length();

        int vc = 0;

        int cc = 0;

        for(int i=0; i<n; i++){

            if((word.charAt(i) == 'a') ||
               (word.charAt(i) == 'e') ||
               (word.charAt(i) == 'i') ||
               (word.charAt(i) == 'o') ||
               (word.charAt(i) == 'u')){

                vc++;

                System.out.println(word.charAt(i));

            }else{

                cc++;

            }

        }

        System.out.println("Number of vowels: "+ vc);
```

```
        System.out.println("Consonants count: "+cc);

    }

}
```

Butterfly pattern

```
import java.util.Scanner;

public class Main{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");

        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {

            for (int j = 1; j <= i; j++) {

                System.out.print("* ");

            }

            for (int j = 1; j <= 2 * (n - i); j++) {

                System.out.print(" ");

            }

            for (int j = 1; j <= i; j++) {

                System.out.print("* ");

            }

            System.out.println();

        }

    }

}
```

```

for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    for (int j = 1; j <= 2 * (n - i); j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
}
}

```

Problem Statement: Given boolean 2D matrix , find the number of islands. A group of connected 1s form an island. For example, the below matrix contains 5 islands.

```

import java.util.*;

public class Main{

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.print("Enter number of rows: ");

        int m=sc.nextInt();
    }
}

```

```

System.out.print("Enter number of columns: ");
int n=sc.nextInt();
int[][] matrix=new int[m][n];
System.out.println("Enter the matrix values (0 or 1): ");
for(int i=0;i<m;i++){
    for(int j=0;j<n;j++){
        matrix[i][j]=sc.nextInt();
    }
}
int count=0;
for(int i=0;i<m;i++){
    for(int j=0;j<n;j++){
        if(matrix[i][j]==1){
            count++;
            dfs(matrix,i,j,m,n);
        }
    }
}
System.out.println("Number of islands: "+count);
}

public static void dfs(int[][] matrix,int i,int j,int m,int n){
    if(i<0||i>=m||j<0||j>=n||matrix[i][j]==0){
        return;
    }
}

```

```

matrix[i][j]=0; // Mark as visited
dfs(matrix,i+1,j,m,n); // Down
dfs(matrix,i-1,j,m,n); // Up
dfs(matrix,i,j+1,m,n); // Right
dfs(matrix,i,j-1,m,n); // Left
dfs(matrix,i+1,j+1,m,n); // Down-Right
dfs(matrix,i+1,j-1,m,n); // Down-Left
dfs(matrix,i-1,j+1,m,n); // Up-Right
dfs(matrix,i-1,j-1,m,n); // Up-Left
}
}

```

Frequency of character

```

import java.util.*;

public class Basic{

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a string:");

        String a = sc.nextLine().toLowerCase();

        int[] arr = new int[26];

        for (int i = 0; i < a.length(); i++) {

            char ch = a.charAt(i);

            if (ch >= 'a' && ch <= 'z') {

                arr[ch - 'a']++;
            }
        }
    }
}

```

```

    }
}
System.out.println("Character\tFrequency");
for (int i = 0; i < 26; i++) {
    if (arr[i] > 0) {
        System.out.println((char) (i + 'a') + "\t\t" + arr[i]);
    }
}
}
}
}

```

Anagram Problem

```

import java.util.Arrays;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
        String str1 = sc.nextLine().replaceAll("\\s", "").toLowerCase();

        System.out.print("Enter second string: ");
        String str2 = sc.nextLine().replaceAll("\\s", "").toLowerCase();
    }
}

```

```

char[] arr1 = str1.toCharArray();
char[] arr2 = str2.toCharArray();

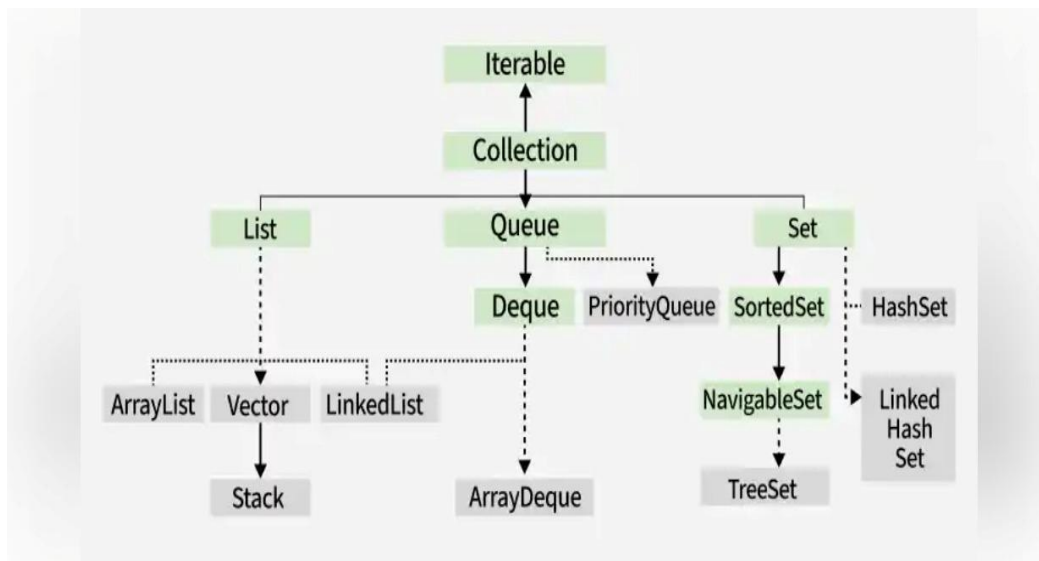
Arrays.sort(arr1);
Arrays.sort(arr2);

if (Arrays.equals(arr1, arr2)) {
    System.out.println("Strings are Anagrams");
} else {
    System.out.println("Strings are Not Anagrams");
}
}
}

```

COLLECTIONS:

1. JAVA COLLECTION FRAME WORK IS A SET OF CLASSES AND INTERFACES THAT PROVIDE REDIMADE DATA STRUCTURES TO STORE AND MANIPULATE GROUP OF OBJECT EFFEICIENTLY
2. Java provide collection interface like list,set,map,with redimates classes such as arraylist,Hastset,HashMap,priorityqueue so you dont have to write data handling code from scratch
3. The collection frame work include productivity by making a code reusable,maintanable and faster to develop



Features Of Collection Framework :

- Provide ready to use data structure
- Offer interfaces(collection,list,set,math,queue)
- To divide standard behaviour
- Support dynamic resizing ,unlike to array with fixed size
- Include algorithms
- Improves code reusabilities and performance

LIST:

Allow duplicate values

SET:

Does not allow duplicate values

ARRAY LIST :

Array list in java is a resizable array provided in java.util package unlike to normal arrays

Array list size can be grow or shrink Dynamically as element or ordered or removed

Elements can be indices similar to array

Duplicates are allowed, default size 10

```
// Creating an ArrayList
ArrayList<Integer> a = new ArrayList<>();
```

```
Main.java :
1 import java.util.ArrayList;
2
3 class Main {
4     public static void main (String[] args) {
5
6         // Creating an ArrayList
7         ArrayList<Integer> a = new ArrayList<>();
8
9         // Adding Element in ArrayList
10        a.add(1);
11        a.add(2);
12        a.add(3);
13        //remove the element in an arrayList
14        a.remove(1);
15        //
16        a.add(1,5);
17        a.add(0,10);
18        //set should replace the element
19        a.set(0,5);
20        // get get the element
21        a.get(1);
22        // Printing ArrayList
23        System.out.println(a);
24        System.out.println(a.get(0));
25        // clear all the values
26        //a.clear();
27
28        a.size();
29        System.out.println(a.size());
30        // check wheather the element is there or not
31        //if there print true else false
32        System.out.println(a.contains(5));
33        // clone it copy the value
34        ArrayList<Integer> b = (ArrayList<Integer>) a.clone();
35        System.out.println(b);
36    }
37 }
```

Output:

```
input
[5, 1, 5, 3]
5
4
true
[5, 1, 5, 3]

...Program finished with exit code 0
Press ENTER to exit console.
```

ARRAYLIST PROBLEM TO FIND SECOND LARGEST NUMBER

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> a = new ArrayList<>();
        a.add(3);
        a.add(2);
        a.add(4);
        a.add(1);
        a.add(5);
        a.add(8);
        a.add(47);
        a.add(19909);
        a.add(33);
        a.add(66);
        a.add(1900000);
        a.add(10);
        int largest = 0;
        int second = 0;

        for(int i = 0; i < a.size()-1; i++) {

            int num = a.get(i);

            if(num > largest) {
                second = largest;
                largest = num;
            }
            else if(num > second && num != largest) {
                second = num;
            }
        }

        System.out.println("Second Largest Element: " + second);
    }
}
```

PATTERN PROBLEM



The screenshot shows a Java IDE with a file named 'Main.java'. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8
9         for (int i = n; i >= 1; i--) {
10             for (int j = n; j >= n - i + 1; j--) {
11                 System.out.print((char)('A' + i - 1));
12             }
13             System.out.println();
14         }
15     }
16 }
```

The output on the right side of the IDE shows the pattern for n=6:

```
6
FFFFF
EEEE
DDDD
CCC
BB
A
...Program finished with exit code 0
Press ENTER to exit console.
```

HASH MAP:

1. HASH MAP IS A PART OF JAVA COLLECTION FRAMEWORK AND IMPLEMENT THE MAP INTERFACES IT STORE THE ELEMENT USE KEY VALUE PAIR WHERE KEY VALUES ARE UNIQUE
2. INTERNALLY USE HASHING HENCE ALLOWS EFFICIENT KEY BASED ACCESS AND INSERT AND REMOVAL
3. STORE THE VALUE WITH THE HELP OF KEYVALUE
4. EVERY KEY VALUE SHOULD BE UNIQUE
5. IT IS NOT A THREAD SPACE , DEFAULT SIZE 16
6. THE CAPACITY OF HASHMAP IS THE NUMBER OF BUCKETS IT CAN HOLD FOR SHORTING ENTERIES

$$\text{NEW CAPACITY} = \text{OLD CAPACITY} * 2;$$

```
// Create a HashMap  
HashMap<String, Integer> hashMap = new HashMap<>();
```

MOST IMPORTANT FUNCTION HASHMAP :

- MAP.GET - TO GET THE VALUES

Map.get(2);//key values

- MAP.CONTAINSKEY(KEY);
- MAP.CONTAINSVALUE(VALUE);
- MAP.REMOVE(KEY VALUE);
- MAP.SIZE(); - LENGTH OF WORD
- MAP.isEmpty(); //create hashmap empty or not
- Map.keySet();
- Map.values();

FUTURE	ARRAY LIST	HASHMAP
STORAGE	SINGLE VALUE	KEY
PACKAGE	JAVA.UTIL.*	JAVA.UTIL.*
ACCESS	INDEX	KEY VALUES
DUPLICATE VALUE	IT ALLOWED	IT ALLOWED
DUPLICATE KEY	KEY IS NOT APPLICABLE	KEY IS NOT ALLOWED
ORDER	INSERTION ORDER MAINTAIN	DOES NOT MAINTAIN
SEARCH	O(N)	O(1) AVERAGE
EXAMPLE	[1 ,2,3]	{ 101 = IMTHI}
NULL KEY VALUE	NO KEY VALUES	ONLY ONCE IN HASHMAP

Sort an Arraylist

```

import java.util.*;

public class Main{

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        ArrayList<Integer> a=new ArrayList<>();

        System.out.print("Enter 10 elements: ");

        for(int i=0;i<10;i++)

            a.add(sc.nextInt());

        Collections.sort(a);

        System.out.println("Sorted ArrayList: "+a);

    }

}

```

Pattern Problem

```

import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        char last = (char)('A' + n - 1);

        for (int i = 0; i < n; i++) {

            for (char j = last; j >= 'A' + i; j--) {

                System.out.print(j);

            }

}

```

```
        System.out.println();
    }
}
}
```

Empty subarray problem

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int count = 0;
        for (int i = 0; i < n; i++) {
            int sum = 0;
            for (int j = i; j < n; j++) {
                sum = sum + arr[j];
                if (sum == 0) {
                    count++;
                }
            }
        }
    }
}
```

```

        }
    }
    System.out.println(count);
}
}

```

PATTERN PROBLEM

```

import java.util.*;

public class Main{

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        for(int i=0;i<n;i++){
            char l=(char)('A'+n-(i+1));
            for(int j=(n-1);j>=i;j--){
                System.out.print((char)(l)+" ");
            }
            System.out.println();
        }
    }
}

```

PATTERN PROBLEM

```

import java.util.*;

public class Main{

    public static void main(String[] args) {

```

```

Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int x=4;
for(int i=0;i<5;i++){
    char l=(char)('A'+x);
    for(int j=(n-1);j>=i;j--)
        System.out.print((char)(l)+" ");
    x--;
    System.out.println();
}
}
}

```

PATTERN PROBLEM

```

import java.util.*;
public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = n - 1; i >= 0; i--) {
            for (int j = 1; j <= i; j++)
                System.out.print("b");
            for (int j = 2; j <= 2 * (n-i); j++)

```



```

        System.out.print("*");
    for (int j = 1; j <= i; j++)
        System.out.print("b");
    System.out.println();

}

}

}

```

Menu list for Biryani Juice Desserts

```

import java.util.*;

public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Welcome to Thaj Hotel\nCustomer is our
family, We serve with passion and LOVEEEEEEE!\nDishes we
make!");

        System.out.println("1. Biriyani served with love");
        System.out.println("2. Juice made with affection");
        System.out.println("3. Dessert crafted with perfection");
        System.out.println("4. Today's Special");
        System.out.println("5. Exit");

        String s = sc.next().toLowerCase();
        if(s.equals("biriyani")||s.equals("1")){

```

```
        System.out.printf("Chicken Biryani \t\t Rs.100\n");
        System.out.printf("Mutton Biryani \t\t Rs.150\n");
        System.out.printf("Prawn Biryani \t\t Rs.170\n");
    System.out.printf("Prawn Biryani \t\t Rs.170\n");
        System.out.printf("Veg Biryani \t\t Rs.70\n");
    System.out.printf("Exit");
    int a[]={100,150,170,170,70};
    }
    else if(s.equals("juice")||s.equals("2")){
        System.out.printf("Watermelon Juice \t\t Rs.40\n");
        System.out.printf("Apple Juice \t\t Rs.50\n");
        System.out.printf("Orange Juice \t\t Rs.40\n");
        System.out.printf("Lemon Jucie \t\t Rs.30\n");
        System.out.printf("Soda \t\t Rs.50\n");
    }
    else if(s.equals("dessert")||s.equals("3")){
        System.out.printf("Ice-cream \t\t Rs.50\n");
        System.out.printf("Milk-shake \t\t Rs.60\n");
        System.out.printf("Waffle \t\t Rs.100\n");
        System.out.printf("Kunafa \t\t Rs.200\n");
        System.out.printf("Cheese Cake \t\t Rs.70\n");
        System.out.printf("Brownie \t\t Rs.50\n");
    }
    else if(s.equals("exit")||s.equals("5")){
```

```
        System.out.println("Thank you we expect you to visit again");
    }

    else{
        System.out.println("Not Available");
    }
}
}
```